2024

(Session : 2023-26)

(Paper ID : 12006)

Time : 3 hours

Full Marks : 80

Note: *Candidates are required to give their answers in their own words as far as practicable.*
The questions are of equal value.
Answer any five questions.

1. Given U (Universal set) $= \{1,2,3,4,5,6,7,8,9\}$, $A = \{1,2,3,4,5\}$, $B = \{3,4,5,7\}$ and $C = \{2,4,5\}$.

Find :

a. $(A-B)\cup(B-A)$
b. $(A\cap B)\cup C$
c. $A^c - B^c$
d. $(A\cup B)-(A\cap B)$

2. Define composition of relations. If $A = \{1,2,3,4\}$, let R be a relation such that $XRY:\ \text{if } Y = X+1$ and S be the relation such that XRY : if $X < Y$. Then find $R\circ S$ and $S\circ R$ with the help of diagram.

3. Define equivalence relation with an example.

4. Let f and g be functions from R to R defined by $f(x) = ax+b, g(x) = 1-x+x^2$. If $(g\circ f)(x) = 9x^2-9x+7$, determine a,b.

5. If f is a function from R to R, where R is a set of real number, defined by $f(x) = \sqrt{2x+1}$, then find the inverse function of f at point $0, 1.5, -25$ and 40.

6. List all partition of $\{a,b,c,d\}$.

7. If $S = \{2, 3, 6, 12, 24, 36\}$, let R be a relation on set S defined by $R = \{(x, y) : x$ divides $y\}$.

a. Construct Hasse diagram

b. Is POSET a Lattice

8. Define group. Prove that set of integers with respect to addition is a commutative group.

9. Define the following with an example :

a. Regular graph

b. Complete graph

c. Euler cycle

d. Trees

10. Define the following :

a. Power set

b. Lattice

c. Ring

d. Operations on set

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